X-Linked Intellectual Disability–Cerebellar Hypoplasia Syndrome (OPHN1 / Oligophrenin-1 Syndrome)

Comprehensive Disease Characteristics Report Target: MONDO:0010337 · OMIM:300486 · **Causal gene:** ***OPHN1*** **(oligophrenin-1), Xq12** Evidence sources are tagged as **[Human clinical]**, **[Mouse]**, **[in vitro/iPSC]**, or **[Computational/aggregated]**.

Summary / Direct Answer

X-Linked Intellectual Disability–Cerebellar Hypoplasia Syndrome is a rare X-linked recessive neurodevelopmental disorder caused by **loss-of-function variants in *OPHN1*** (Xq12), which encodes **oligophrenin-1, a 91-kDa Rho-GTPase-activating protein (RhoGAP)** highly expressed in fetal brain (Billuart et al., 1998, *Nature*,).

P 9582072

Loss of oligophrenin-1's GAP activity removes negative control of RhoA, causing **constitutive RhoA/ROCK/MLC2 pathway hyperactivation** plus brain-region–specific **cAMP/PKA dysregulation**, which impairs dendritic spine maturation, AMPA-receptor recycling and synaptic-vesicle endocytosis. Clinically this produces intellectual disability with early hypotonia, motor/speech delay, behavioral problems, seizures, strabismus, and the hallmark neuroimaging triad of **cerebellar vermis hypoplasia, ventriculomegaly, and fronto-temporal atrophy** (Bergmann et al., 2003,).

P 12805098

The disorder is non-degenerative and compatible with long survival; no approved disease-modifying therapy exists, but **the ROCK/PKA inhibitor fasudil reverses behavioral and anatomical deficits in adult Ophn1 mice** (Meziane et al., 2016,),

P 27146843

establishing a mechanism-based therapeutic rationale.

1. Disease Information

Overview. A monogenic (Mendelian) syndromic form of X-linked intellectual disability (XLID). Core features: non-progressive intellectual disability (mild–severe), neonatal hypotonia, delayed motor/speech milestones, behavioral disturbance (hyperactivity, attention problems), epilepsy, ocular abnormalities (strabismus, nystagmus), and distinctive brain MRI showing cerebellar (vermis) hypoplasia with enlarged lateral ventricles. Originally described as a cause of *nonspecific* XLID, it was later recognized as a distinct, recognizable syndrome once brain imaging was systematically applied (Bergmann 2003, ; P 12805098 Zanni 2005,). P 16221952

[Human clinical]

Key identifiers. - **MONDO:** 0010337 - **OMIM:** 300486 (Intellectual developmental disorder, X-linked syndromic, Billuart type / "Mental retardation, X-linked, with cerebellar hypoplasia and distinctive facial appearance") - **Gene OMIM:** *OPHN1* 300127 · **HGNC:** 8183 · **NCBI Gene:** 4983 · **Ensembl:** ENSG00000079482 · **UniProt:** O60890 - **Orphanet:** OPHN1 syndrome (ORPHA grouping under non-specific/syndromic XLID) - **ICD-10:** Q04.3 (other reduction deformities of brain) + F70–F79 (intellectual disability) as applicable; **ICD-11:** LD90.Y / 6A00 (disorders of intellectual development) with imaging-based coding - **MeSH:** "Mental Retardation, X-Linked"; "Intellectual Disability"; "Cerebellar Diseases"

Synonyms / alternative names. OPHN1 syndrome; Oligophrenin-1 syndrome; X-linked intellectual developmental disorder, Billuart type; MRX60 (historical non-syndromic designation); X-linked mental retardation with cerebellar hypoplasia and distinctive facial appearance.

Information source. Derived from aggregated disease-level resources (OMIM, Orphanet, HPO) and individual-patient case reports/small cohorts; there is no large EHR-based registry for this ultra-rare disorder.

Phenotypic spectrum note. Beyond classic syndromic/non-syndromic XLID, rare *OPHN1* variants have also been reported in **autism spectrum disorder** and **(childhood-onset) schizophrenia** cohorts

(P 20479760 ; P 30378261 "Variants of this gene have been also identified in autism spectrum disorder and schizophrenia,"), P 27146843 consistent with a shared synaptic-plasticity pathophysiology. **[Human clinical]**

---

2. Etiology

Primary cause — genetic. Hemizygous loss-of-function variants in *OPHN1* in males. "Here we report the characterization of a new gene on the long arm of the X-chromosome (position Xq12) and the identification in unrelated individuals of different mutations that are predicted to cause a loss of function" (Billuart 1998, P 9582072).
[Human clinical]

Genetic risk factors. - **Causal variants:** nonsense, frameshift (small indels), splice-site, whole/partial-gene and exonic deletions (e.g., exon 19 frameshift deletion, P 12805098 ; exon 21 deletion, P 32872024 ; exons 21–22 deletion, P 18261018 ; intragenic deletions, P 30534410), and rarer non-disruptive in-frame/missense variants affecting the BAR or C-terminal proline-rich domains (P 33638601

).
P 24105372
- **Dosage sensitivity:** a Xq12q13.1 **duplication** encompassing *OPHN1* also produced syndromic MR
( ),
P 18512229
indicating sensitivity to both loss and gain of copy number. - **Sex:** male sex is the principal risk factor (X-linked recessive); heterozygous females are usually unaffected carriers.

Environmental risk factors. None established; this is a monogenic disorder. No toxin, infectious, dietary, or occupational exposure is implicated as causal.

Protective factors. In carrier females, **favorably skewed X-inactivation** (preferential silencing of the mutant X) is protective against manifestation; conversely, skewing toward the mutant allele can unmask disease in females
( ).
P 23416624
No protective pharmacologic/dietary factors are known. **[Human clinical]**

Gene–environment interactions. Not applicable/none documented.

---

3. Phenotypes

Onset is **congenital/neonatal** (hypotonia) with developmental features emerging in infancy–early childhood; the cognitive deficit is **stable (non-progressive)**, while epilepsy and behavior can **fluctuate**. Severity is **highly variable, even within families**
(
P 33638601
).
P 29510240
Frequencies below are qualitative estimates aggregated from case series.

Phenotype	Type	HPO term	Frequency	Notes
Intellectual disability (mild→severe)	cognitive/ behavioral	HP:0001249	~100%	Defining feature; variable severity
Neonatal hypotonia	clinical sign	HP:0001319 / HP:0008935	Common	Early presenting sign
Motor delay	sign	HP:0001270	Very common	—
Speech/language delay	sign	HP:0000750	Very common	—
Cerebellar vermis hypoplasia	imaging/ physical	HP:0001320 / HP:0002335	Common (not obligate)	Hallmark; absent in a subset (P 29510240)
Ventriculomegaly / enlarged lateral ventricles	imaging	HP:0002119 / HP:0006956	Common	Often rostral/ frontal predominant
Fronto-temporal atrophy	imaging	HP:0012156-related	Frequent	P 12805098 P 24105372
Seizures (incl. myoclonic-astatic)	sign	HP:0001250 / HP:0200134	Frequent	P 12805098
Strabismus	sign	HP:0000486	Frequent	—
Nystagmus	sign	HP:0000639	Occasional	P 32872024
Ataxia / impaired coordination	sign	HP:0001251	Frequent	Cerebellar origin
Behavioral problems (hyperactivity, frustration intolerance)	behavioral	HP:0000752 / HP:0000708	Common	P 24105372
Hypogenitalism / genital anomalies	physical	HP:0000811	Subset	P 12805098

Phenotype	Type	HPO term	Frequency	Notes
Facial dysmorphism (prominent nasal root, deep-set eyes, long face)	physical	HP:0000271	Common, subtle	P 24105372
Strain on adaptive function (ADL dependence)	QoL	—	Severity-dependent	Lifelong support frequently required

> Quote: "The distinctive phenotype comprised moderate to severe mental retardation, myoclonic-astatic epilepsy, ataxia, strabismus and hypogenitalism." (P 12805098)
> **[Human clinical]**

Quality-of-life impact. Lifelong intellectual disability affects education, independent living and employment; epilepsy and behavioral/attention problems add caregiver burden. No disease-specific EQ-5D/SF-36/PROMIS datasets exist; impact is inferred from the ID + epilepsy + behavior phenotype.

---

4. Genetic / Molecular Information

- **Causal gene:** *OPHN1* (HGNC:8183; NCBI Gene 4983; OMIM 300127), Xq12, ~26 exons, encoding oligophrenin-1 (UniProt O60890; 802 aa; ~91 kDa). Protein domains: N-terminal **BAR** domain, PH domain, central **RhoGAP** domain, C-terminal **proline-rich** region.
- **Variant classes (ACMG):** predominantly **pathogenic/likely-pathogenic loss-of-function** — nonsense, frameshift, canonical splice-site, and multi-exon/whole-gene **deletions**. Non-disruptive in-frame deletions/missense variants (BAR or proline-rich domains) are rarer and pose VUS interpretation challenges. "the identified genomic variants involve the region encoding the central RhoGAP domain ... predicted to result in a complete loss of function" (P 33638601).

- **Functional consequence: loss of function** (loss of RhoGAP catalytic activity → failure to terminate RhoA signaling). Copy-number **gain** (duplication) is also pathogenic (P 18512229).
- **Origin: germline**; inherited from carrier mothers or *de novo*; **somatic mosaicism** reported (a missense variant in mosaic state, P 29510240). Not a somatic/cancer gene.
- **Allele frequency:** pathogenic variants are absent/ultra-rare in gnomAD; *OPHN1* shows constraint against LoF. **[Computational/aggregated]**
- **Modifier genes:** none firmly established. The main modifier of female expression is the **X-inactivation ratio** (epigenetic), not a trans-acting gene (P 23416624).
- **Epigenetic/post-transcriptional:** *OPHN1* mRNA undergoes **A-to-I RNA editing (ADAR2)** and **alternative splicing** during human brain development, peaking at gestational weeks 20–33 — relevant to normal regulation and potentially to phenotype modulation (Barresi 2014, P 24637888). **[Human/in vitro]**
- **Chromosomal abnormalities:** intragenic and contiguous Xq12–q13.1 deletions and one duplication; detectable by array-CGH/MLPA (P 18261018 P 18512229 P 30534410 P 32872024).

5. Environmental Information

No environmental, lifestyle, or infectious contributors are implicated — the disease is purely monogenic. Environmental factors, diet, smoking/alcohol, toxins, radiation, and pathogens are **not applicable** as causes. (Standard prenatal/perinatal neuroprotective care applies generically but is not disease-specific.)

---

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Loss-of-function variant in *OPHN1*** (nonsense/frameshift/splice/deletion) **results in** absent or non-functional oligophrenin-1 protein (loss of RhoGAP activity). [Demonstrated — P 9582072]
2. Loss of RhoGAP activity **leads to** failure to accelerate GTP hydrolysis on RhoA → **constitutive RhoA activation**. [Demonstrated/inferred from GAP biochemistry — P 9582072]
3. Constitutive RhoA **leads to ROCK (Rho-kinase) hyperactivation**, confirmed in OPHN1-deficient human iPSC neurons where ROCK remains elevated throughout differentiation. [Demonstrated — P 27160703]
4. ROCK hyperactivation **results in** increased phosphorylation of myosin phosphatase target subunit (MYPT1) and MLC2 → **excess actomyosin contractility and dysregulated actin cytoskeleton**. [Demonstrated — P 27160703 P 35563851]
5. **Branch A (postsynaptic):** cytoskeletal dysregulation + loss of OPHN1–Homer1b/c and OPHN1–endophilin A1 interactions **lead to** mispositioned spine endocytic zone, **impaired AMPA-receptor recycling**, and **failure of dendritic spine maturation** (excess immature filopodia). [Demonstrated —

P 24966368

P 19481455

P 17728457]

6. *Sub-branch (activity-dependent plasticity):* group-I mGluR activation normally triggers **rapid dendritic OPHN1 translation** (mGluR1-dependent, **FMRP-independent**) required for **mGluR-LTD** and the associated persistent reduction of surface AMPARs via endophilin A2/3; loss of OPHN1 **results in** impaired protein-synthesis-dependent LTD — a role separable from its basal-synaptic Rho-GAP/Homer1b/c function. [Demonstrated —

P 22017989]

7. **Branch B (presynaptic):** OPHN1 loss **leads to** brain-region-specific **cAMP/PKA pathway dysregulation** and defective synaptic-vesicle endocytosis, abolishing **PKA-dependent presynaptic plasticity** in hippocampus and amygdala. [Demonstrated —

P 24298161

P 19481455]

8. Combined pre/postsynaptic dysfunction **results in altered short-term plasticity** (reduced paired-pulse facilitation) and impaired synapse maturation without neuronal loss. [Demonstrated, mouse —

P 17728457]

9. During development, failure of Rho-GTPase–dependent control of neurite/dendrite outgrowth and migration **contributes to cerebellar (vermis) hypoplasia** and **ventricular enlargement**; neuropathology shows **Purkinje-cell heterotopia** in cerebellar white matter. [Demonstrated, human fetus —

P 23416624]

10. These circuit- and structural-level defects **manifest clinically as** intellectual disability, hypotonia, ataxia, epilepsy, strabismus, and behavioral disturbance. [Human clinical —

P 12805098

P 16221952]

Detail by category

- **Molecular pathways:** RhoA → ROCK → MYPT1/MLC2 actomyosin pathway (KEGG hsa04810 regulation of actin cytoskeleton; Rho GTPase cycle, Reactome R-HSA-194840); cAMP/PKA signaling (presynaptic). GO: **GO:0007265** Ras/Rho signal transduction; **GO:0043087** regulation of GTPase activity; **GO:0051056** regulation of small GTPase signal transduction.
- **Cellular processes:** dendritic spine morphogenesis (**GO:0060997**), regulation of synaptic plasticity (**GO:0048167**), **long-term synaptic depression (GO:0060292)** / mGluR-dependent LTD, **protein-synthesis-dependent plasticity** (rapid dendritic OPHN1 translation; P 22017989), synaptic vesicle endocytosis (**GO:0048488**), receptor recycling (**GO:0001881**), actin cytoskeleton organization (**GO:0030036**), neuron projection development (**GO:0031175**). No apoptosis/neurodegeneration — this is a **developmental/connectivity** disorder, not a neurodegenerative one.
- **Protein dysfunction:** loss of RhoGAP catalytic function (not misfolding/aggregation); some in-frame variants retain stable but hypofunctional protein affecting BAR-domain membrane curvature/hippocampal roles (P 24105372).
- **Metabolic / immune / infectious involvement:** none implicated.
- **Molecular profiling:** iPSC-neuron studies show elevated ROCK activity and reduced NR4A1 (neurite-outgrowth nuclear receptor), normalized by fasudil (P 27160703); developmental RNA editing/splicing of *OPHN1* (P 24637888). No disease-specific large-scale transcriptomic/proteomic/metabolomic datasets published.
- **Functional genomics:** modeled by *Ophn1* knockout/conditional mice and OPHN1-knockdown/knockout iPSC neurons (below).

Upstream vs downstream: *OPHN1* LoF → RhoA/ROCK hyperactivation are **upstream**; spine/vesicle/plasticity defects and cerebellar hypoplasia are **downstream**. Cell types (CL): neurons (**CL:0000540**), pyramidal/hippocampal neurons (**CL:0000598**), cerebellar Purkinje cells (**CL:0000121**), glutamatergic neurons (**CL:0000679**).

7. Anatomical Structures Affected

- **Organ/system:** central nervous system (nervous system, **UBERON:0001016**); principally brain (**UBERON:0000955**).
- **Primary sites: cerebellum** (UBERON:0002037), especially **cerebellar vermis** (UBERON:0004086) — hypoplasia/lower-vermian agenesis; **cerebral lateral ventricles** (UBERON:0002285) — enlargement; **frontal and temporal lobes** (UBERON:0016525 / UBERON:0001871) — atrophy; **hippocampus** (UBERON:0002421) — alterations with certain BAR-domain variants (P 24105372); **cisterna magna** — cystic dilatation.
- **Secondary involvement:** visual axis (strabismus/nystagmus); genitourinary (hypogenitalism in a subset).
- **Tissue/cell level:** neural tissue; affected cell populations include **cortical/hippocampal pyramidal neurons** and **cerebellar Purkinje cells** (heterotopic in white matter, P 23416624).
- **Subcellular compartments (GO CC): dendritic spine (GO:0043197)**, **postsynaptic density (GO:0014069)**, **presynaptic active zone / synaptic vesicle (GO:0008021)**, **endocytic zone / clathrin-coated pit**, **actin cytoskeleton (GO:0015629)**.
- **Lateralization:** brain involvement is generally **bilateral**, though **asymmetric cerebellar hypoplasia** has been described (P 12805098).

8. Temporal Development

- **Onset:** congenital/neonatal (hypotonia at birth; some structural anomalies detectable prenatally on fetal MRI/ultrasound, P 23416624).
  Pattern: **chronic, insidious** developmental presentation.

- **Progression:** the intellectual disability is **static/non-progressive** (a neurodevelopmental, not neurodegenerative, condition). Epilepsy may evolve and behavioral issues fluctuate; cerebellar/ventricular findings are stable structural features. Disease duration: **chronic, lifelong**.
- **Remission:** no spontaneous remission of ID; seizures/behavior may improve with treatment.
- **Critical period:** synaptic and cerebellar development (fetal–early postnatal) is the window of structural vulnerability. Importantly, mouse data show **functional deficits remain pharmacologically reversible in adulthood** (P 27146843), implying a therapeutic window beyond early development. **[Mouse]**

9. Inheritance and Population

- **Inheritance: X-linked recessive** (OMIM 300486). Affected individuals are predominantly **hemizygous males**; **carrier females** are usually unaffected but can manifest with **skewed X-inactivation** (P 23416624).
- **Penetrance:** essentially **complete in hemizygous males**; **expressivity is highly variable**, even among siblings sharing the same variant (P 33638601). P 29510240
- **Germline mosaicism / de novo:** both inherited and *de novo* variants occur; **somatic mosaicism** has been documented (P 29510240).
- **Anticipation:** not applicable (not a repeat-expansion disorder).
- **Founder effects / consanguinity:** no founder mutation; consanguinity not a driver (X-linked). Recurrent large deletions arise independently.
- **Carrier/allele frequency:** pathogenic alleles ultra-rare; absent from population controls (gnomAD).

- **Epidemiology: ultra-rare**; precise prevalence/incidence undefined (Orphanet: prevalence unknown). Background context: nonspecific XLMR affects ~0.15–0.3% of males, of which *OPHN1* accounts for ~1%, rising to ~12% among males with ID **plus** cerebellar anomalies (P 9582072). (P 16221952) Only on the order of dozens of families reported worldwide.
- **Sex ratio:** strongly **male-predominant**; rare symptomatic females with skewed XCI or structural variants.
- **Geographic distribution:** worldwide, no endemic clustering; reported across European, Brazilian, Japanese and other populations (P 24105372). (P 30534410)

10. Diagnostics

- **Recommended approach:** clinical suspicion (male with ID + hypotonia + cerebellar/ventricular MRI findings ± epilepsy/strabismus) → **brain MRI** → **molecular genetic testing**.
- **Imaging (key diagnostic): brain MRI** demonstrating cerebellar vermis hypoplasia, enlarged (often rostral) lateral ventricles, fronto-temporal atrophy, and cisterna magna dilatation (P 12805098). (P 24105372) Fetal MRI/ultrasound may detect ventriculomegaly and vermian hypoplasia prenatally (P 23416624). (RadLex imaging features; no disease-specific biomarker on labs.)
- **Genetic testing:**
- **Exome/genome sequencing (WES/WGS)** and **XLID / epilepsy–ID multigene panels** including *OPHN1* are first-line for undiagnosed ID; "We emphasize the power of next

generation techniques (X-exome sequencing, whole-exome sequencing and targeted multi-gene panel) to expand the phenotypic and mutational spectrum of OPHN1-related ID" (P 29510240).

- **Single-gene sequencing** of *OPHN1* where phenotype is classic.
- **Chromosomal microarray (CMA)/array-CGH** and **MLPA** to detect intragenic/whole-gene **deletions and duplications**, which are a common mutation class (P 18261018 P 18512229 P 30534410 P 32872024). "we suggest that MLPA analysis should be performed as an alternative diagnostic test" (P 32872024).
- **Laboratory tests/biomarkers:** no specific blood/urine biochemical marker; routine labs normal. Karyotype usually normal. mtDNA/repeat-expansion testing **not applicable**.
- **Electrophysiology:** EEG for seizure characterization (e.g., myoclonic-astatic epilepsy).
- **Histopathology (research/autopsy):** cerebellar Purkinje-cell heterotopia (P 23416624).
- **Clinical criteria / differential diagnosis:** no formal diagnostic criteria; diagnosis is molecular. Differentials include other XLID-with-cerebellar-hypoplasia genes (e.g., **CASK**, **OPHN1 vs. ATRX**, **ARX**), Dandy-Walker/pontocerebellar hypoplasias, and syndromic ID with posterior-fossa anomalies; distinguish by gene, facial gestalt, and MRI pattern (P 16506132 P 30534410).
- **Screening:** no population newborn screening (not actionable/biochemical). **Cascade carrier testing** of at-risk female relatives once the familial variant is known; prenatal/preimplantation testing available for known variants.

11. Outcome / Prognosis

- **Survival/mortality: not life-limiting per se**; the condition is compatible with normal or near-normal lifespan. No disease-specific mortality rate established; risks relate to complications of severe epilepsy or profound disability (generic).
- **Morbidity/disability: lifelong intellectual disability** is the dominant morbidity, with variable needs for educational, communication, mobility and behavioral support; epilepsy and strabismus add morbidity. ICF-level impairments in cognition, communication, and adaptive functioning.
- **Disease course:** static cognitive deficit; seizures and behavior may be modifiable with treatment. Recovery of baseline cognition is not expected; early intervention improves functional outcomes (generic rehabilitation evidence).
- **Prognostic factors:** severity correlates loosely with overall phenotype but there is **no robust genotype–phenotype correlation**, limiting individual prognostication (P 33638601). Presence/severity of epilepsy and degree of ID are the main functional predictors.
- **Prognostic biomarkers:** none validated.

12. Treatment

No approved disease-modifying therapy exists; management is symptomatic and supportive.

- **Supportive/rehabilitative (standard of care):** special education, **speech/language therapy** (NCIT:C15193), **physical therapy** (NCIT:C15327) and **occupational therapy** (NCIT:C15335) for hypotonia/motor delay, behavioral/psychological support. Ophthalmologic management of strabismus (corrective lenses/surgery).
- **Pharmacotherapy (symptomatic): antiseizure medications** for epilepsy (agent chosen by seizure type, e.g., valproate/levetiracetam; NCIT:C264 anticonvulsant class); stimulants/ behavioral agents for ADHD-type symptoms as clinically indicated. Pharmacogenomics: no disorder-specific PGx guidance; standard ASM PGx (e.g., HLA-B for carbamazepine) applies generically.

- **Mechanism-based experimental therapy — ROCK/PKA inhibition (fasudil):** the leading translational candidate. "chronic treatment in adult mouse with Fasudil, is able to counteract vertical and horizontal hyperactivities, restores recognition memory and limits the brain ventricular dilatation observed in Ophn1" (P 27146843); fasudil "fully rescued" hippocampus/amygdala learning deficits (P 24298161) and corrected OPHN1-deficient iPSC neurons (P 27160703). CHEBI: fasudil (CHEBI:75172); target NCIT: Rho-associated kinase inhibitor. **Human trials specific to OPHN1 are not yet reported. [Mouse / in vitro]**
- **Gene/cell/RNA-based therapy:** none in clinical development; conceptually plausible given defined LoF mechanism, but no vectors/ASOs reported for *OPHN1*.
- **Surgical/interventional:** none disease-specific (standard epilepsy or strabismus surgery as indicated).
- **Treatment strategy:** multidisciplinary (neurology, genetics, developmental pediatrics, ophthalmology, rehabilitation); personalized to seizure type and developmental needs. Combination (pathway inhibitor + symptomatic care) is the envisaged future model.

13. Prevention

- **Primary prevention:** not possible (genetic); risk reduction centers on **reproductive counseling**.
- **Genetic counseling:** carrier mothers have **50% transmission risk** to each son (affected) and 50% of daughters are carriers (X-linked recessive). **Cascade testing** of maternal relatives once the variant is identified.
- **Secondary prevention (reproductive options for known-variant families): prenatal diagnosis** (CVS/amniocentesis) and **preimplantation genetic testing (PGT-M)**; prenatal imaging may detect ventriculomegaly/vermian hypoplasia.
- **Tertiary prevention:** seizure control, early developmental intervention, behavioral and physical therapies to limit secondary complications and maximize function.

- **Immunization / public-health / environmental interventions:** not applicable.

14. Other Species / Natural Disease

- **Orthologous gene:** mouse ***Ophn1*** (NCBI Gene 94190; MGI); rat *Ophn1* (NCBI Gene 309989). Protein is highly conserved; RhoGAP function conserved across vertebrates.
- **Taxonomy of experimental hosts:** *Mus musculus* (NCBITaxon:10090); *Homo sapiens* (NCBITaxon:9606) iPSC-derived neurons.
- **Natural disease in other species: no naturally occurring OPHN1 disease reported** in companion animals or wildlife (OMIA: none established); veterinary relevance is limited to engineered models.
- **Comparative biology:** mouse *Ophn1* KO recapitulates ventricular enlargement, spatial-memory and social deficits, hyperactivity, and dendritic-spine immaturity, confirming evolutionary conservation of the RhoGAP–synapse mechanism
(P 17728457).
- **Transmission/zoonosis:** not applicable (genetic disorder).

15. Model Organisms

- **Mouse (primary mammalian model):** constitutive and **conditional *Ophn1* knockouts** (Mus musculus). Phenotype recapitulation: enlarged lateral ventricles, impaired spatial memory, impaired social behavior, hyperactivity, reduced paired-pulse facilitation, and reduced mature dendritic spines — "we generated a mouse model of ophn1 deficiency and showed that it mimics the ventricles enlargement without affecting the cerebellum morphoanatomy"
(P 17728457).
Model limitation: the mouse does **not** reproduce the human **cerebellar hypoplasia**, an important divergence. Models also revealed presynaptic cAMP/PKA deficits
(P 24298161)
and enabled **fasudil rescue** proof-of-concept

(P 27146843).

[Mouse]

- **Human iPSC-derived neurons (in vitro cellular model):** OPHN1 loss-of-function iPSC neurons show sustained ROCK hyperactivity, impaired neuronal differentiation/morphology, and rescue by fasudil/ROCK inhibition (P 27160703). Useful for human-specific neurodevelopmental mechanisms and drug testing. **[in vitro]**
- **Primary neuronal cultures / rat hippocampus:** used to dissect OPHN1–Homer1b/c (AMPAR recycling; P 24966368) and OPHN1–endophilin A1 (SV endocytosis; P 19481455) functions, and postnatal developmental changes in synaptic transmission/plasticity (P 35563851).
- **Model types available:** knockout, conditional/floxed (region- and stage-specific), knockdown; patient-derived iPSC lines. **Resources:** MGI (mouse), IMPC/IMSR, Cellosaurus (iPSC lines).
- **Research applications:** synaptic plasticity mechanisms, dendritic spine biology, Rho-GTPase signaling in cognition, and preclinical ROCK-inhibitor pharmacology.

Supported vs. Refuted Hypotheses

Supported: - *OPHN1* LoF causes the disorder via **RhoA/ROCK hyperactivation** (P 9582072). P 27160703 ✔ - Disease reflects **synaptic dysfunction (pre- and postsynaptic)** rather than neurodegeneration (P 17728457

P 24298161

P 19481455

).

P 24966368

✔ - **Cerebellar hypoplasia is variable/non-obligate**; phenotype spans non-syndromic to syndromic

(

P 29510240

).

P 33638601

✔ - **ROCK/PKA inhibition (fasudil) reverses deficits in adulthood** in models

(

P 27146843

P 24298161

).

P 27160703

✔

Refuted / not supported: - That OPHN1 causes *purely* non-syndromic MR — refuted once imaging revealed consistent cerebellar/ventricular signatures

(

P 12805098

P 16221952

).

P 24105372

- A tight genotype–phenotype correlation — not supported; expressivity varies even within families

(

).

P 33638601

- An environmental/infectious contribution — none found; disorder is monogenic.

Limitations and Future Directions

- **Ultra-rare** disorder: epidemiologic parameters (precise prevalence, incidence, natural-history cohorts, QoL metrics) are undefined.

- **No human therapeutic trials**; fasudil evidence is preclinical (mouse/iPSC) — translation and a human treatment window remain to be established.
- **Model gap:** mouse lacks cerebellar hypoplasia, limiting study of that key human feature; new models (e.g., cerebellar organoids) are needed.
- **Variant interpretation** of non-disruptive (BAR/proline-rich, missense) variants is challenging and would benefit from standardized functional assays.
- Roles of **RNA editing/alternative splicing** and **X-inactivation** as modifiers warrant further study.

Key References (PMID)

9582072 (Billuart 1998, *Nature*, gene discovery) · 12805098 (Bergmann 2003, recognizable syndrome) · 16221952 (Zanni 2005, cohort/frequency) · 17728457 (Khelfaoui 2007, KO mouse) · 19481455 (Nakano-Kobayashi 2009, endophilin A1) · 22017989 (Nadif Kasri 2011, mGluR-LTD / rapid OPHN1 synthesis) · 24966368 (Nakano-Kobayashi 2014, Homer1b/c) · 20479760 (Piton 2011, ASD/SCZ X-linked synaptic genes) · 30378261 (Ambalavanan 2019, childhood-onset schizophrenia) · 24298161 (Khelfaoui 2014, cAMP/PKA, fasudil) · 27146843 (Meziane 2016, fasudil adult rescue) · 27160703 (Compagnucci 2016, iPSC/ROCK) · 24637888 (Barresi 2014, RNA editing/splicing) · 24105372 (Santos-Rebouças 2014, BAR domain/hippocampus) · 23416624 (Rocas 2013, female fetus/XCI/Purkinje heterotopia) · 29510240 (Moortgat 2018, no cerebellar anomaly subset) · 33638601 (Nuovo 2021, allelic spectrum) · 32872024 (Bogliş 2020, MLPA/deletion) · 18512229 (Bedeschi 2008, duplication) · 18261018 (Madrigal 2008, deletion/aCGH) · 30534410 (Iida 2019, Dandy-Walker/deletion) · 35563851 (Cresto 2022, postnatal synaptic development) · 16506132 (Guillén-Navarro 2006, XLMR review).
